Synoptic LE: Tree Fluxes 2022

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1 Read in and collate all the files in the Processed Data folder

2 Remove bad fluxes and calculate how many there were

```
##Remove any samples with Flags in the final qaqc_flag columns
final_dat <- all_dat %>%
  filter(!grepl("TRUE", CH4_rejected_model, ignore.case = TRUE)) %>%
  filter(!grepl("TRUE", CO2_flux_rejected , ignore.case = TRUE))

##Removed both CH4 and CO2 obs when CO2 fluxes were negative and when visual checks showed w
```

```r
# count before and after filtering
n_total_co2  <- nrow(all_dat)
n_total_ch4  <- nrow(all_dat)

final_dat <- all_dat %>%
  filter(CH4_rejected_model == FALSE | is.na(CH4_rejected_model)) %>%
  filter(CO2_flux_rejected  == FALSE | is.na(CO2_flux_rejected))

n_final      <- nrow(final_dat)
n_removed_co2 <- sum(all_dat$CO2_flux_rejected  == TRUE, na.rm = TRUE)
n_removed_ch4 <- sum(all_dat$CH4_rejected_model == TRUE, na.rm = TRUE)
n_removed_either <- n_total_co2 - n_final

# build summary table
filter_summary <- tibble::tibble(
  Metric = c(
    "Total observations (joined dataset)",
    "CO2 - observations rejected (n)",
    "CO2 - observations rejected (%)",
    "CH4 - observations rejected (n)",
    "CH4 - observations rejected (%)",
    "Removed by both flags simultaneously (n)",
    "Total unique observations removed (n)",
    "Total unique observations removed (%)",
    "Final observations retained (n)"
  ),
  Value = c(
    n_total_co2,
    n_removed_co2,
    round(100 * n_removed_co2    / n_total_co2, 1),
    n_removed_ch4,
    round(100 * n_removed_ch4    / n_total_co2, 1),
    sum(all_dat$CO2_flux_rejected == TRUE &
        all_dat$CH4_rejected_model == TRUE, na.rm = TRUE),
    n_removed_either,
    round(100 * n_removed_either / n_total_co2, 1),
    n_final
  )
)

filter_summary %>%
  kable(
```

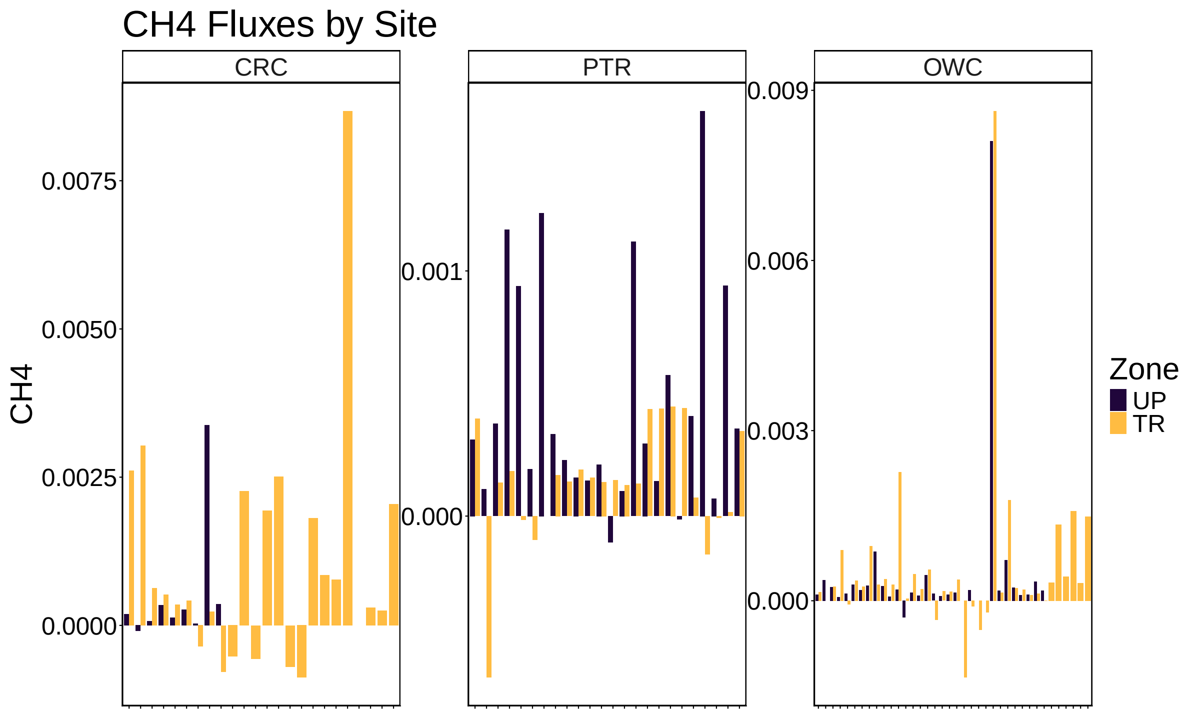
```r
  caption  = "Post-Join QA/QC Filtering Summary",
  align    = c("l", "r"),
  booktabs = TRUE,
  format   = "latex"
) %>%
kable_styling(
  latex_options = c("striped", "hold_position"),
  full_width    = FALSE
) %>%
pack_rows("CO2 flags",          2, 3) %>%
pack_rows("CH4 flags",          4, 5) %>%
pack_rows("Combined filtering", 6, 9)
```

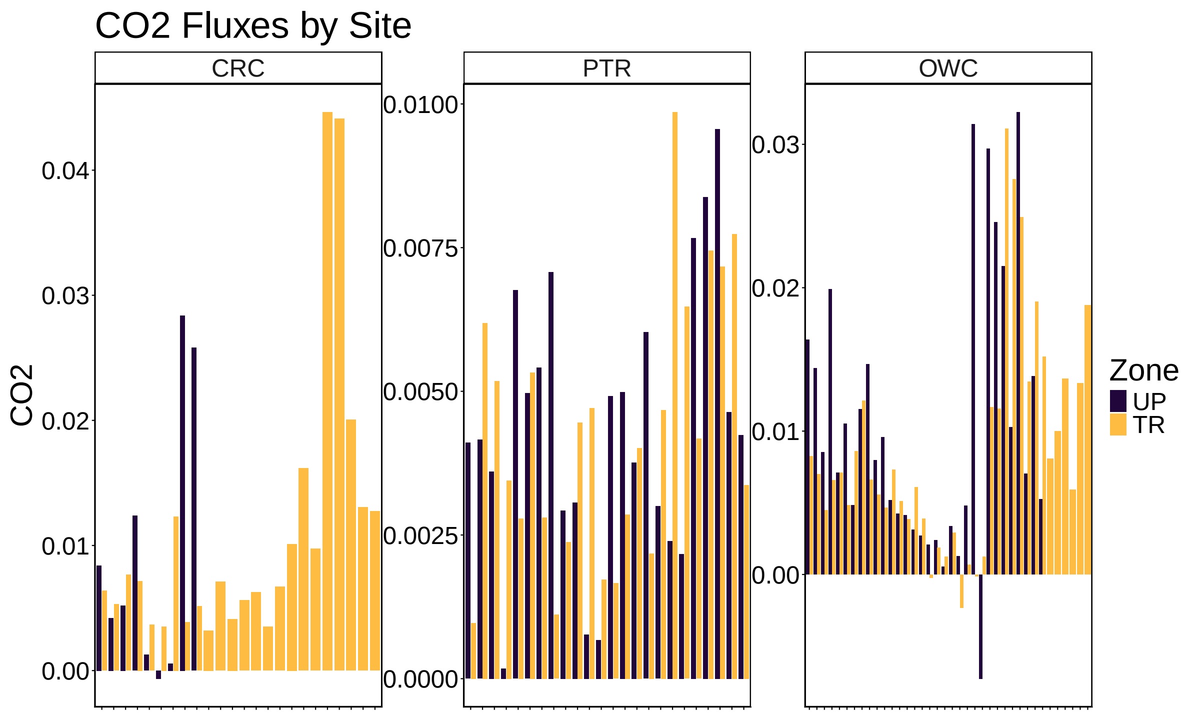
Table 1: Post-Join QA/QC Filtering Summary

Metric	Value
Total observations (joined dataset)	151.0
CO2 flags	
CO2 — observations rejected (n)	7.0
CO2 — observations rejected (%)	4.6
CH4 flags	
CH4 — observations rejected (n)	0.0
CH4 — observations rejected (%)	0.0
Combined filtering	
Removed by both flags simultaneously (n)	0.0
Total unique observations removed (n)	7.0
Total unique observations removed (%)	4.6
Final observations retained (n)	144.0

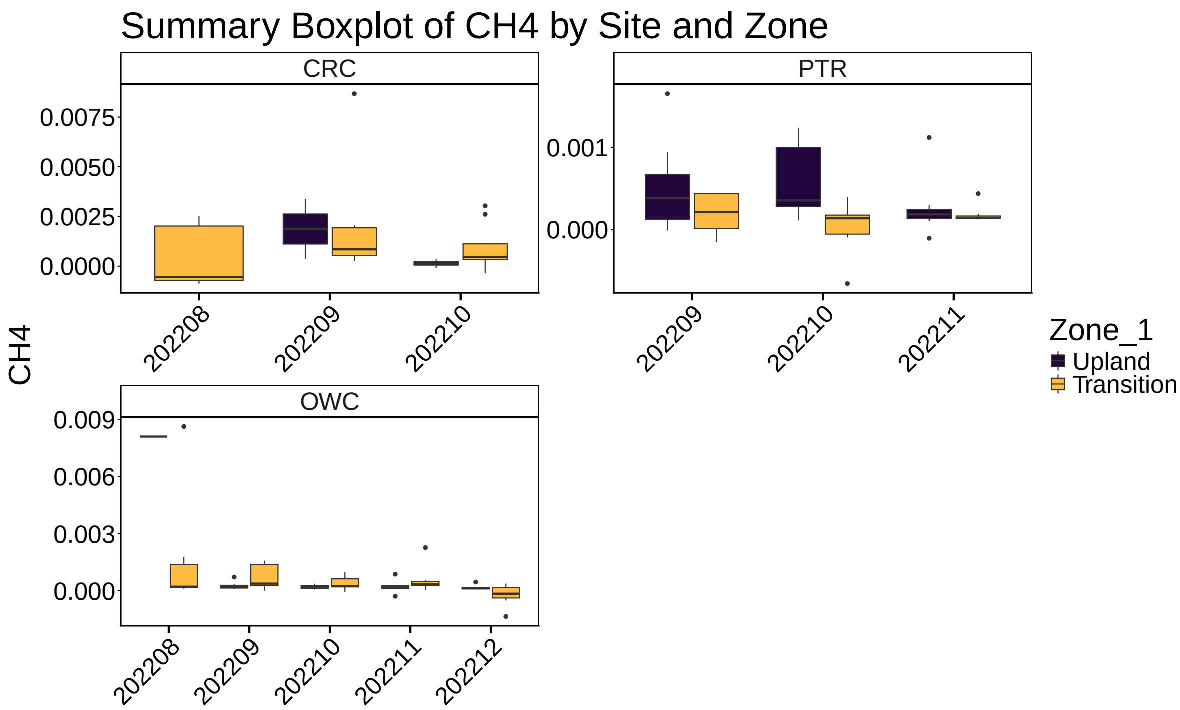
3 Write out files

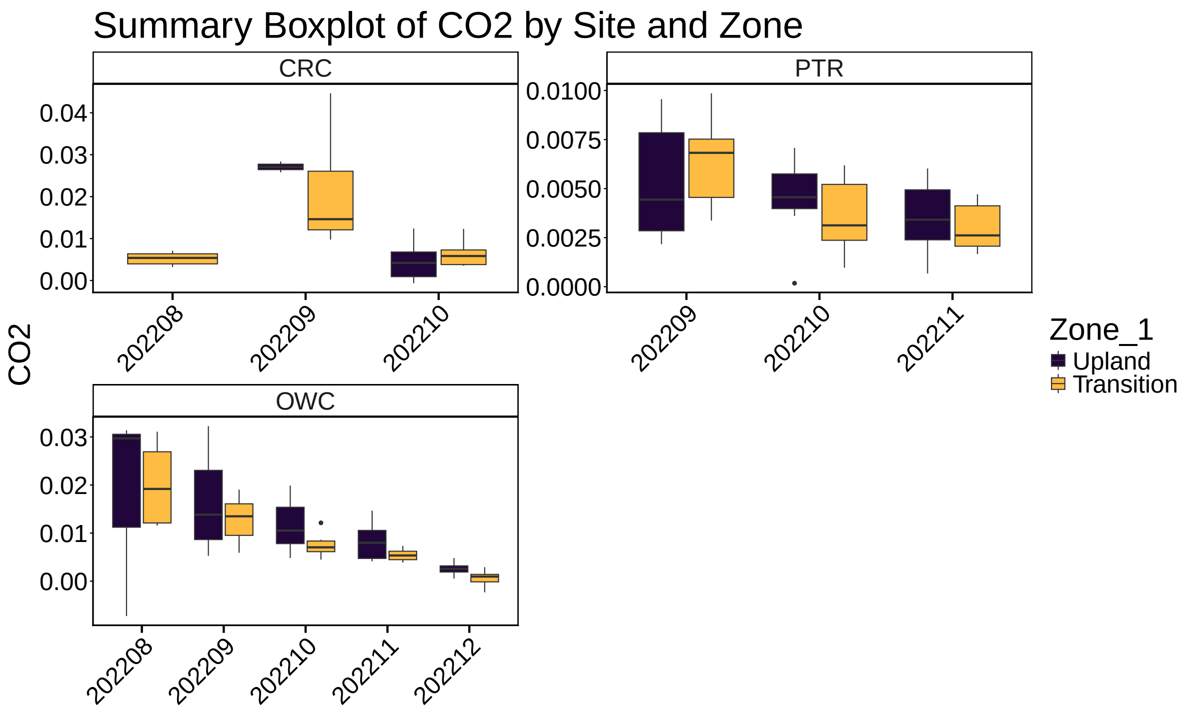
4 All Data by Year, Site and Zone




CO2 Fluxes by Site
CRC
PTR
OWC
CO2
0.00
0.01
0.02
0.03
0.04
0.0000
0.0025
0.0050
0.0075
0.0100
0.00
0.01
0.02
0.03
Zone
UP
TR

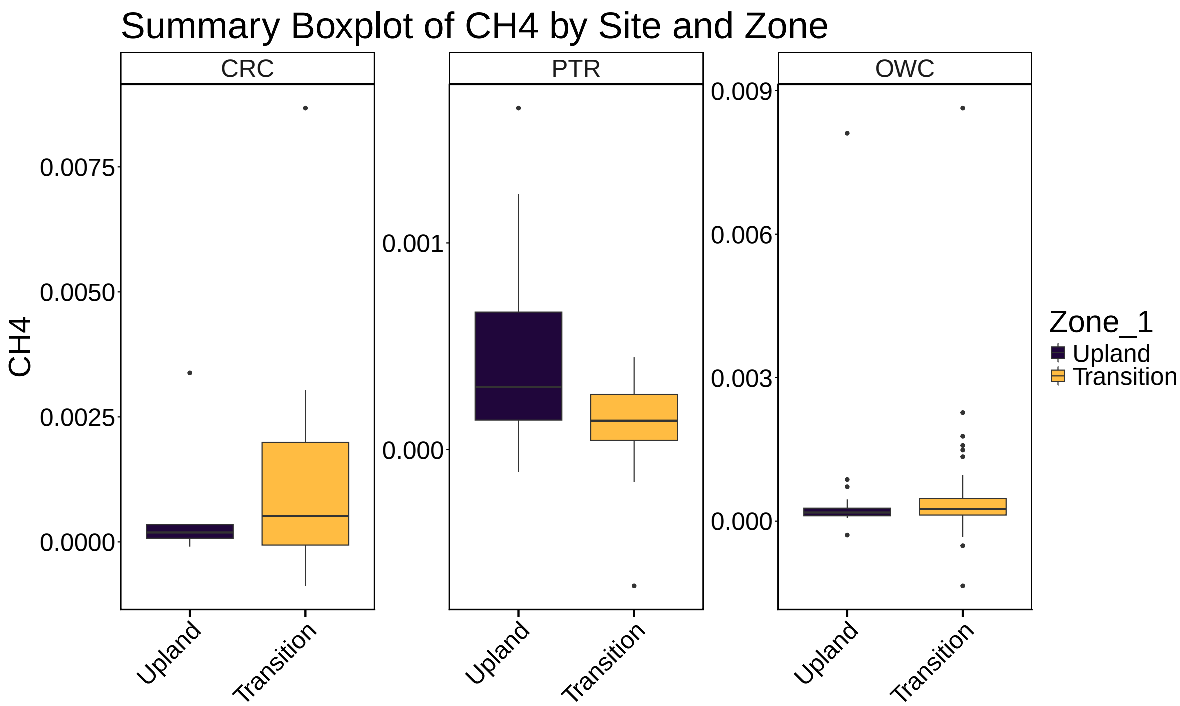
5 Boxplot summary by Date

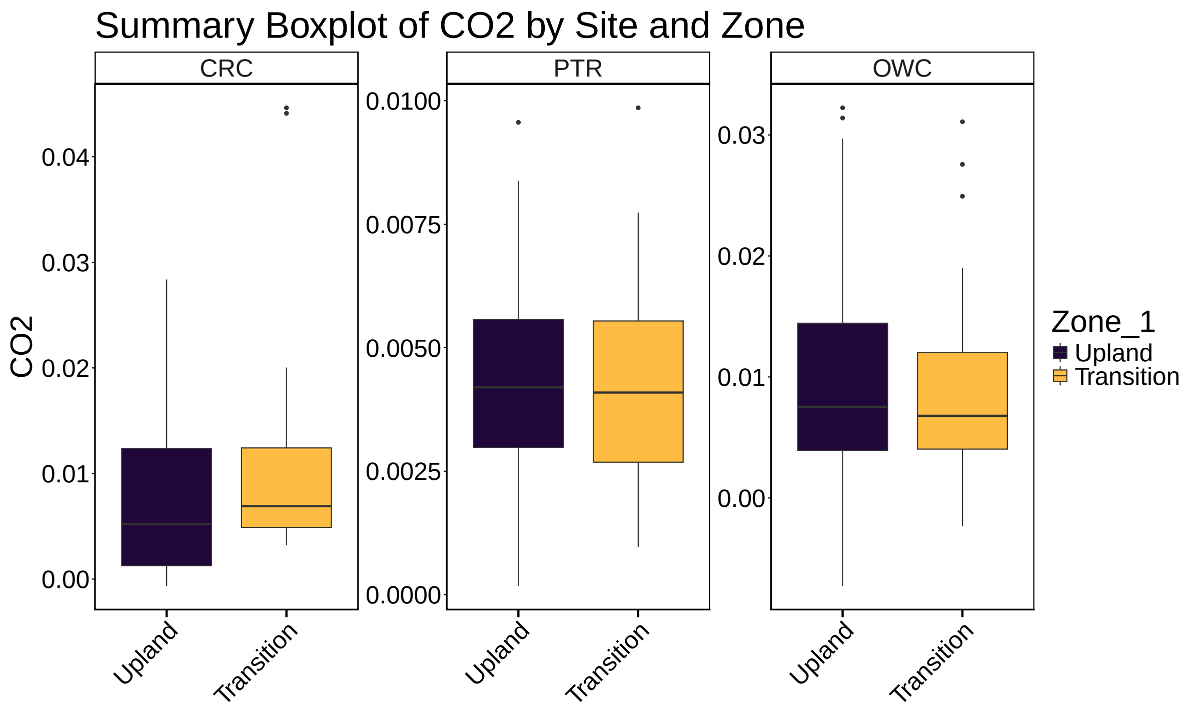




Summary Boxplot of CO2 by Site and Zone
CRC
PTR
OWC
CO2
0.04
0.03
0.02
0.01
0.00
0.0100
0.0075
0.0050
0.0025
0.0000
202208
202209
202210
202211
202212
Zone_1
Upland
Transition

6 Boxplot summary by Site





7 Calculate summary metrics

8 Summary Tables

Table 2: Summary Statistics

Site	Zone_1	min_ch4	max_ch4	mean_ch4	min_co2	max_co2	mean_co2
CRC	Upland	-0.0000939	0.0033799	0.0005201	-0.0006479	0.0283806	0.0095048
CRC	Transition	-0.0008791	0.0086760	0.0011037	0.0032105	0.0446401	0.0109227
PTR	Upland	-0.0001068	0.0016524	0.0004560	0.0001774	0.0095626	0.0043942
PTR	Transition	-0.0006586	0.0004471	0.0001382	0.0009688	0.0098576	0.0042789
OWC	Upland	-0.0002913	0.0081095	0.0004853	-0.0072671	0.0322457	0.0102418
OWC	Transition	-0.0013540	0.0086358	0.0005990	-0.0023300	0.0310917	0.0087207